

1 — Prof's Blueprint (High Priority)

Exam: MCQ + written, **30 pts total**. Practical (worked-number) problems dominate; fewer theory questions. Calculator allowed; **no programming** required. One A4 sheet allowed (this one). **Highest-frequency tools:** $\text{dec} \leftrightarrow \text{bin}$, two's complement, IEEE 754, gcd, mod arithmetic + tables, inverses/zero divisors, poly-mod reduction, group axioms/order/Lagrange, FLT big powers, expression trees + Lisp lists, substitution vs rewrite.

2 — Sets & Exact vs Approximate

$\mathbb{N}$: here $\{1, 2, 3, \dots\}$ (**no 0!**); $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$; $\mathbb{Q} = \{p/q : p \in \mathbb{Z}, q \neq 0\}$; $\mathbb{R}; \mathbb{Z}[x]$ = polynomials with integer coefficients. **Prime:** integer > 1 with exactly 2 positive divisors (1, itself). **Composite:** > 1 , more than 2 divisors. **1 is neither;** 2 is the only even prime. **Exact** = symbolic: fractions kept as reduced pairs, radicals/ π kept symbolically, arbitrary precision. **Approximate** = floats: rounding errors accumulate; floats **cannot** represent 0.1 exactly (repeating binary). Exact CAS features: no round-off, results provably correct, may be slow/huge.

3 — Integers in Memory (Two's Complement)

Unsigned n bits: range $0 \dots 2^n - 1$ (max = $2^n - 1$).

Signed n bits: range $-2^{n-1} \dots 2^{n-1} - 1$ (one extra negative); MSB = 1 means negative. **Negate** = flip bits + 1: -62 (7-bit): 0111110 $\rightarrow$ 1000001 $\rightarrow$ 1000010₂. **Back:** MSB 1 $\rightarrow$ negate: 1111011 $\rightarrow$ 0000101 = 5 $\rightarrow$ -5. $-2^{n-1} = 100 \dots 0$ has no positive counterpart.

Bits needed for $-m$: smallest n with $2^{n-1} \geq m$: -200 $\rightarrow n=9$ ($2^8=256$), -300 $\rightarrow n=10$, -1030 $\rightarrow n=12$, $-2 \times 10^9 \rightarrow n=32$. **Min/max in n bits:** 8-bit: unsigned 0..255, signed -128..127; 16-bit signed: -32768..32767; 32-bit: $\pm 2^{31}$ bounds. **Arithmetic:** add = bitwise with carry (1010₂ + 0111₂ = 10001₂ = 17); multiply = shift-and-add (1010₂ $\times$ 11₂ = 11110₂ = 30). Shifts: $x \ll k = x \cdot 2^k$, $x \gg k = x \text{ div } 2^k$ (integer only).

Dec $\leftrightarrow$ Bin

Int ($\div 2$, read remainders bottom-up): 21₁₀ = 10101₂; 33₁₀ = 10001₂; 26₁₀ = 11010₂; 205₁₀ = 11001101₂. **Bin $\rightarrow$ dec:** 1011₂ = 8 + 2 + 1 = 11. **Frac ($\times 2$, take the integer part each step):** 0.6875 $\rightarrow$ 0.1011₂; 0.1₁₀ = 0.00011... (periodic); 0.1011₂ = $\frac{1}{2} + \frac{1}{8} + \frac{1}{16} = \frac{11}{16} = 0.6875$. **Mixed:** 21.09375 = 10101.00011₂ (0.09375 = $\frac{1}{16} + \frac{1}{32} = \frac{3}{32}$).

4 — IEEE 754 Floats

Normalized: $(-1)^s(1.f)_2 2^e$, stored with bias: value = $(-1)^s(1.f)_2 2^{E-\text{bias}}$, bias = 127 (single)/1023 (double). Layout: **float 1+8+23**, **double 1+11+52** bits. **Convert dec $\rightarrow$ float:** dec $\rightarrow$ bin $\rightarrow$ normalize to $1.f \times 2^e$ $\rightarrow$ store s , fraction bits (drop leading 1), $E = e + \text{bias}$. **Worked:** 21.09375 = 1.010100011₂ $\times 2^4 \rightarrow s=0$, $f=010100011$, $e=4$; single-precision $E = 4 + 127 = 131 = 10000011_2$. **Bin $\rightarrow$ dec:** sum 2^{-k} place values: 0.1011₂ = 0.6875. **Zero:** cannot arise as $1.f \times 2^e$ ($1.f \neq 0$); IEEE reserves all-0 exponent bits for +0, -0. $\times/\div$ by 2 = exponent shift; 0.1 + 0.2 gives rounding error.

Special bit patterns: ± 0 : exponent = mantissa = 0 (sign bit only); subnormal: exponent = 0, mantissa $\neq 0$; $\pm \infty$: exponent = all 1s, mantissa = 0; NaN: exponent = all 1s, mantissa $\neq 0$.

Range Qs (bias = 24, 16 bits: 1+5+10, $E \geq 0$): largest = $1.99902 \times 2^{31-24} = 255.875$; smallest positive = $1.0 \times 2^{-24} \approx 5.96046 \times 10^{-8}$; negatives mirror.

5 — GCD, Fractions, Coprime

Division: $a = bq + r$, $0 \leq r < b$ ($q = a \text{ div } b$, $r = a \text{ mod } b$). **Euclid:** $\text{gcd}(a, b) = \text{gcd}(b, a \text{ mod } b)$, $\text{gcd}(a, 0) = a$; gcd = last nonzero remainder. $\text{gcd}(126, 84) = 42$; $\text{gcd}(125, 100) = 25$; $\text{gcd}(144, 60) = 12$; $\text{gcd}(48, 18) = 6$. **Coprime** $\Leftrightarrow \text{gcd}(a, b) = 1$. **Fraction = pair (p, q)**, reduced: divide by $g = \text{gcd}(p, q)$: $\frac{6}{12} \rightarrow (1, 2)$; $\frac{125}{100} \rightarrow (5, 4)$. Arithmetic: $\frac{p_1}{q_1} + \frac{p_2}{q_2} = (p_1q_2 + p_2q_1, q_1q_2)$; product = (p_1p_2, q_1q_2) . **Rational number $\frac{p}{q}$** = pair with $\text{gcd}(a, b) = 1$; rationals are terminating or repeating decimals ($\frac{1}{3} = 0.(3)$, $\frac{9}{11} = 0.(81)$); irrationals ($\sqrt{2}, \pi, e$) never repeat.

6 — Arbitrary-Precision Integers (APN)

Fixed n bits $\rightarrow$ max $\approx 2^{n-1}$; to do exact big arithmetic, store number as a **digit list in base B** , each entry $\in [0, B-1]$, **least significant first**:

$$x = d_0 + d_1B + d_2B^2 + \dots \Leftrightarrow [d_0, d_1, d_2, \dots]$$

1,234,567,890 with $B=10^3 \Rightarrow [890, 567, 234, 1]$. **Addition:** elementwise with carry. $x=1,234,567$, $y=987,654$, $B=10^3$: $[567, 234, 1] + [654, 987, 0] \Rightarrow [221, 222, 2] \Rightarrow [2, 222, 221]$. **List $\rightarrow$ dec:** $\sum d_i B^i$; **dec $\rightarrow$ list:** repeated $\div B$. Complexity: add $O(n)$, schoolbook mult $O(n^2)$, FFT mult $O(n \log n)$.

7 — Prime Factorization

FTA: every $n > 1$ has **unique** prime factorization; multiply factorized numbers by **adding exponents**: $75 = 3^1 5^2$, $360 = 2^3 3^2 5$ $\Rightarrow 75 \times 360 = 2^3 3^3 5^3 = 27000$. List form $[p, e]$: $360 = [(2, 3), (3, 2), (5, 1)]$. **Caveat:** factoring large n is hard (no known poly-time algorithm); $\sim N/\ln N$ primes $\leq N$.

8 — CAS Quirks (Contagious Reals)

CAS keep irrationals symbolic: $\sqrt{2} \rightarrow \sqrt{2}$, but $\sqrt{2.0} \rightarrow 1.41421$; one float anywhere makes the whole expression numeric: $\ln 2 + \ln 3 \rightarrow \ln 6$ (exact), $\ln 2 + \ln(3.0) \rightarrow 1.79176$. Simplify radicals by rewrite rule $\sqrt{k^2 a} = k\sqrt{a}$: $\sqrt{8} = \sqrt{4 \cdot 2} = 2\sqrt{2}$. Maxima: expand, factor, ratsimp, subst, solve, makelist, first/rest/append/map; :lisp \$expr shows Lisp form (e.g. $a+b \rightarrow (\text{MPLUS SIMP } \$A \$B)$).

9 — Expression Trees

Internal nodes = operators, leaves = operands (variables/constants). Unary $(-x)$, binary $(a+b)$, ternary $(c/a : b)$. **Construction:** root = **lowest-precedence** operator; recurse on operands. **Precedence** (low $\rightarrow$ high): $+, -, \times, \div, \wedge$. **Associativity** fixes shape for repeated ops: **Left-assoc** ($-, \div, \text{mod}, \ll$): $a-b-c = (a-b)-c$. **Right-assoc** (power, assignment $=$): $2^3^2 = 2^9 = 512$; $a=b=c$ groups $a = (b=c)$. e.g. $a+b \cdot c + d^2$: root +, left child a , right child $+(b \cdot c, d^2)$. $a^{b^c} = d^{f^g \cdot k} = (a^{b^c} - d^e) - (f^g \cdot k)$ ($-$ left-assoc). **Lisp lists** = prefix: $a+b \rightarrow (+ a b)$. **Traversals** (DFS): pre/in/post-order print prefix/infix/postfix. Unary subtrees padded with NULL node for uniform in-order DFS. **Worked (custom precedence/assoc):** $\odot$ (low, left), $\bullet$ (mid, left), $\odot$ (high, right): $a \odot b \bullet c \odot d \odot f \odot g \Rightarrow a \odot ((b \bullet c) \bullet d) \odot (e \odot (f \odot g))$; root is $\odot$. Multi-operand: $(+ 1 2 3) \rightarrow ((+ 1 2) 3) \rightarrow 1 + 2 + 3$.

10 — Transformations & Rewrite Rules

Substitution (value/symbol level): replace a symbol/subtree by another, everywhere: $a+b \rightarrow t$ in $(a+b)^2 + 2(a+b+c) \Rightarrow t^2 + 2(t+c)$. **Expansion = distributivity:** $a(b+c+d) \rightarrow ab+ac+ad$; $(a+b)(c+d+e) \rightarrow ac+bc+ad+bd+ae+be$; $(a_1 \dots a_n)^p \rightarrow a_1^p \dots a_n^p$; set version $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. **Rewrite rule:** pattern $\rightarrow$ replacement, applied bottom-up to smallest subexpressions first until stable. **Needs types** (rule matches structure and leaf types): notation $\text{symbol} :: \text{type}$, e.g. $5 :: \text{integer}$, $x :: \text{variable}$. $\sqrt{k^2 a} = k\sqrt{a}$ needs $k :: \text{integer}$; $\frac{p}{q} \rightarrow \frac{p/g}{q/g}$, $g = \text{gcd}(p, q)$, $p, q :: \text{integer}$, $q \neq 0$. **Substitution is value-level; rewrite is structure-level.**

11 — Modular Arithmetic

$a \equiv b \pmod{n} \Leftrightarrow n \mid (a-b) \Leftrightarrow$ same remainder mod n . **Residue class:** $[a]_n = \{a + kn : k \in \mathbb{Z}\}$; representative $\in [0, n-1]$; $\mathbb{Z}_n = \{[0], \dots, [n-1]\}$. **Operations well-defined:** $[a] + [b] = [a+b]$, $[a][b] = [ab]$, $[a]^p = ([a])^p$. Reduce then compute; never form huge powers. **Negatives:** add multiples of n : $-5 \equiv 2 \pmod{7}$, $-1234 \equiv 8 \pmod{9}$ ($-1234 + 138 \cdot 9 = 8$); $143 \equiv 0 \pmod{13}$, $1234 \equiv 2 \pmod{8}$. **Notation conversion:** $[8]_{11} + [9]_{11} = [6]_{11} \Leftrightarrow 8 + 9 \equiv 6 \pmod{11}$; $17 + 9 \equiv 2 \pmod{8}$ $\Leftrightarrow [17]_8 + [9]_8 = [2]_8$. **Exponentiation:** $[7]_8^5 = [2]_8^5 = [256] = [1]_8$; $2^{10} \equiv 4 \pmod{5}$; $[26]_5 + [17]_5 = [43]_5 = [3]_5$; $[7]_3 \cdot [11]_3 = [77]_3 = [2]_3$.

Inverses & Zero Divisors

Additive inverse of $[a]$ is $[-a]$ (always exists). **Multiplicative inverse** $[a]^{-1}$ exists $\Leftrightarrow \text{gcd}(a, n) = 1$ (Bézout); in tables, **row a has a 1 $\Leftrightarrow a$ invertible**. **Zero divisors:** $[a], [b] \neq [0]$ with $[a][b] = [0]$; exist $\Leftrightarrow n$ composite. **Mod 5** (prime): $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$.

	$\times$	1	2	3	4		$\times$	1	2	3	4
	1	1	2	3	4		1	1	2	3	4
mod 5:	2	2	4	1	3	mod 4:	2	2	0	2	
	3	3	1	4	2		3	3	2	1	
	4	4	3	2	1						

Mod 4 (composite): 2^{-1} **does not exist**; $[2]_4[2]_4 = [0]_4$ (zero divisor, row 2 lacks 1).

12 — Polynomial Modular Arithmetic

$[p(x)]_m(x) = p(x) + m(x)\mathbb{Z}[x] = \{p + mw : w \in \mathbb{Z}[x]\}$; classes = **residues of division by m** , degree $< \deg m$. Reduce: divide p by m (or use $\equiv$ rules); **mod x :** keep only constant term $[x^2 + 5x - 11]_x = [-11]_x$, $[x^5 - 3x^3 + 8]_x = [8]_x$. **mod $(x^2 + 1)$:** use $x^2 \equiv -1$; classes are $a + bx$. $[x^3 + 2x + 1] = [x + 1]$ (since $x^3 = x \cdot x^2 \equiv -x$). Squaring: $(a + bx)^2 \equiv (a^2 - b^2) + 2abx$, e.g. $(9 - x)^2 \equiv 80 - 18x$. **Powers:** $[p]^k = [p^k]$, reduce at end. Isomorphisms: $\{[p]_x\} \cong \mathbb{Z}$; $\{[p]_{x^2+1}\} \cong \mathbb{Z}x + \mathbb{Z}$ (Gaussian integers $a + bi$, $a, b \in \mathbb{Z}$).

13 — Groups

Group $(S, \circ)$: 4 axioms — **closure** ($a, b \in S \Rightarrow a \circ b \in S$), **associativity** ($((a \circ b) \circ c) = a \circ (b \circ c)$), **identity** e (unique), **inverses** (unique). Additive: identity 0; multiplicative: identity 1. **Verify all four.**

Check all four axioms — closure is the one people forget. Non-groups: $(\mathbb{Z}, \cdot)$ fails inverses ($2^{-1} = \frac{1}{2} \notin \mathbb{Z}$); $(\mathbb{N}, +)$ fails inverses ($-2 \notin \mathbb{N}$); $(\mathbb{Q}, \cdot)$ fails (0); $(\mathbb{Z}_6, *)$ fails. $(\mathbb{Z}, +)$ and $(\mathbb{Z}_n, +)$ are groups ($e=0$, inverse $[-a]$). $\mathbb{Z}_n^* = \{[1], \dots, [n-1]\}$ under $\times$ is a group $\Leftrightarrow n$ prime; e.g. $(\mathbb{Z}_5^*, \times)$: $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$. $\mathbb{Z}_4^* = \{1, 2, 3\}$ is **not** a group: $2 \cdot 2 = 0 \notin$ set (closure fails). **4-axiom check for $\mathbb{Z}_5^*$:** (1) closure: $[2][4] = [3] \in$ set, etc. (all products in $\{1, 2, 3, 4\}$); (2) identity $[1]$; (3) associativity inherited from integers (needs $[abc] = a[b]c$); (4) every element has inverse (table). $\Rightarrow$ group.

Order, Cyclic, Lagrange

Order of group $|G| = \# \text{elements}$; $|(\mathbb{Z}_n, +)| = n$, $|(\mathbb{Z}_p^*, \times)| = p-1$. **Order of element** $\text{ord}(g)$ = smallest $n > 0$ with $g^n = e$ (additive: $n \cdot g = 0$); $\text{ord}(e) = 1$; if none, ∞ . **Lagrange:** G finite $\Rightarrow \text{ord}(g) \mid |G|$; corollary $g^{|G|} = e$. **Generators:** G cyclic $\Leftrightarrow$ some g has $\text{ord}(g) = |G|$; then $G = \langle g \rangle$. In $\mathbb{Z}_n$: generators are $[k]$ with $\text{gcd}(k, n) = 1$ (e.g. $\mathbb{Z}_4^+$: $[1], [3]$; $\mathbb{Z}_8^+$: $[1], [3], [5], [7]$). **Worked:** $\mathbb{Z}_5^*$: $\text{ord}(1)=1$, $\text{ord}(4)=2$, $\text{ord}(2)=\text{ord}(3)=4$; generators 2, 3. $\mathbb{Z}_8^*$: $\text{ord}[2]=4$, $\text{ord}[7]=8$. $\mathbb{Z}_{13}^*$: $|G|=12$, 2 is a generator ($2^{12} \equiv 1$, no smaller power $\equiv 1$; powers hit all 12 elements). **Possible orders** divide $|G|$: $|\mathbb{Z}_{16}^+|=16 \Rightarrow$ orders $\in \{1, 2, 4, 8, 16\}$; $|\mathbb{Z}_{11}^*|=10 \Rightarrow$ orders $\in \{1, 2, 5, 10\}$.

Multiplication Table: D_3 (triangle symmetries)

$D_3 = \{e, r, r^2, \ell_1, \ell_2, \ell_3\}$, $|D_3| = 6$: $r = \text{rot } 120^\circ$, $r^2 = \text{rot } 240^\circ$, $\ell_i = \text{reflections}$. Entry = apply **column first, then row**: $r \circ r = r^2$, $\ell_3 \circ \ell_1 = r^2$, $\ell_2 \circ \ell_1 = r$, $\ell_3 \circ \ell_2 = r$. Inverses: $r^{-1} = r^2$, $(r^2)^{-1} = r$, $\ell_i^{-1} = \ell_i$.

	e	r	r^2	ℓ_1	ℓ_2	ℓ_3
e	e	r	r^2	ℓ_1	ℓ_2	ℓ_3
r	r	r^2	e	ℓ_2	ℓ_3	ℓ_1
r^2	r^2	e	r	ℓ_3	ℓ_1	ℓ_2
ℓ_1	ℓ_1	ℓ_3	ℓ_2	e	r^2	r
ℓ_2	ℓ_2	ℓ_1	ℓ_3	r	e	r^2
ℓ_3	ℓ_3	ℓ_2	ℓ_1	r^2	r	e

14 — Fermat's Little Theorem

Fermat's little theorem: $a^{p-1} \equiv 1 \pmod{p}$ for p prime with $\text{gcd}(a, p) = 1$ ($p \nmid a$); residue form $[a^{p-1}]_p = [1]_p$.

Derives from Lagrange on $\mathbb{Z}_p^*$ ($|\mathbb{Z}_p^*| = p-1$). **Big powers:** reduce base mod p ; write exponent = $(p-1)q + c$; then $a^{(p-1)q+c} \equiv a^c \pmod{p}$. **Worked:** $123^{123} \pmod{7}$: $[123]_7 = [4]_7$, $123 = 6 \cdot 20 + 3 \Rightarrow 4^{6 \cdot 20 + 3} \equiv 4^3 = 64 \equiv 1$. $19^{98} \pmod{7}$: $19 \equiv 5$, $68 = 6 \cdot 11 + 2 \Rightarrow 5^2 = 25 \equiv 4$. $271^{24} \pmod{11}$: $271 \equiv 7$, $24 = 10 \cdot 2 + 4 \Rightarrow 7^4 = 2401 \equiv 3$. $290^{293} \pmod{23}$: $290 \equiv 14$, $293 = 22 \cdot 13 + 7 \Rightarrow 14^7 \equiv 19$. **FLT not applicable:** $2^{16} \pmod{7}$ ($\text{gcd}(2, 7) = 2 \neq 1$), $11^8 \pmod{9}$ (9 not prime). Without FLT: $[16^4]_7 = [2^4]_7 = [16]_7 = [2]_7$; $[10^{100}]_9 = [1]_9$; $[23^2]_9 = [5^2]_9 = [25]_9 = [7]_9$; $[42]_{12} = [6]_{12}$. **Worked with base-reduction only** (no FLT): $4569^{4569} \pmod{11}$: $4569 \equiv 4$, so $4^{4569} = 4^{10 \cdot 456 + 9} \equiv 4^9 \equiv 3$ (track $4^2=16 \equiv 5$, $4^4=25 \equiv 3$, $4^8=9$, $4^9=36 \equiv 3$).

15 — Worked Solutions (Step-by-Step)

1–3. Binary conversions & arithmetic

Dec→bin (unsigned): repeatedly $\div 2$, write quotient q and remainder r ; continue until $q=0$; read remainders **bottom-up**.
 45: $45=2\cdot 22+1$, $22=2\cdot 11+0$, $11=2\cdot 5+1$, $5=2\cdot 2+1$, $2=2\cdot 1+0$, $1=2\cdot 0+1 \Rightarrow 101101_2$.
Check: $32+8+4+1=45$. **Bin→dec:** sum place values $1100101_2=32+16+8+1=57$.

Signed dec→bin: (1) write $|m|$ in n bits (2) **flip bits** (3) $+1$. -37 (8-bit):
 $00100101 \rightarrow 11011010 \rightarrow 11011011$.

Check: 11011011 as unsigned $= 219$;
 $219-256=-37$. **Bin→dec:** MSB 1 $\Rightarrow$ negate:
 $10110001 \Rightarrow 01001110 \Rightarrow 01001111=79 \Rightarrow -79$.
Add (carry): $10110+01111=100101$ ($22+15=37$).
Mult (shift&add):
 $101 \times 101 = 1101 + 110100 = 1000001$ ($13 \times 5 = 65$).
Shift = multiply/divide by 2: $100110 \gg 1 = 10011$ ($38 \rightarrow 19$).

4. IEEE 754 conversion (worked both ways)

Dec→bits: (1) convert integer and fraction to binary (2) normalize to $1.f \times 2^e$ (3) $E=e+127$; write $s|E|f$.
 $6.75: 6=110_2$, $0.75=0.11_2$; $110.11_2=1.1011_2 \times 2^2$;
 $E=2+127=129=10000001_2 \Rightarrow$ bits
 $0\ 10000001\ 101100000000000000000000$.
 $-3.75: 11.11_2=1.111_2 \times 2^1$, $s=1$,
 $E=128=10000000_2 \Rightarrow$ bits
 $1\ 10000000\ 111000000000000000000000$.
Bits→dec: read s, E, f ; $e=E-127$; value
 $= (-1)^s (1.f)_2 \times 2^e$.
 $0\ 10000010\ 011000 \dots: e=130-127=3$,
 $(1.011)_2=1.375$, so $1.375 \times 2^3 = 11$.
Traps: exponent all-0s $\Rightarrow$ denormal (no hidden 1); all-1s $\Rightarrow \infty/\text{NaN}$.

5. GCD via Euclid

Method: replace the bigger by $a-bq$ ($= a \bmod b$) until 0; **last nonzero remainder = gcd**.
 $\text{gcd}(96, 36): 96=3\cdot 36+24$, $36=2\cdot 24+12$, $24=2\cdot 12+0 \Rightarrow \text{gcd}=12$. **Check:** $96/12=8$, $36/12=3$.
Coprime $\iff \text{gcd}=1$: $\text{gcd}(21, 22)=1$ (consecutive) $\Rightarrow$ coprime; $\text{gcd}(14, 21)=7 \neq 1$ not.
Reduce fraction: $\frac{24}{36} = \frac{24/12}{36/12} = \frac{2}{3}$.

6. Arbitrary precision (base B) numbers

Dec→list: repeated $\div B$; remainders are digits (LSB-first). $58,432$ with $B=1000$: $r=432$, $q=58$; then $r=58$, $q=0 \Rightarrow [432, 58]$. **Check:** $58\cdot 1000+432=58,432$.
List→dec: $[890, 567, 234, 1]=1\cdot 10^9+234\cdot 10^6+567\cdot 10^3+890=1,234,567,890$.
Add (carry c): $[567, 234, 1] + [654, 987, 0]$: col0 $567+654=1221 \rightarrow$ digit 221, $c=1$; col1 $234+987+1=1222 \rightarrow 222$, $c=1$; col2 $1+0+1=2$. Result $[221, 222, 2]=2,222,221$.

7. Min/max in n bits

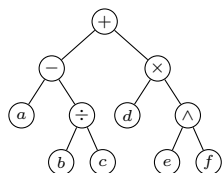
Unsigned: $0 \dots 2^n - 1$ (all n bits free). Signed: $-2^{n-1} \dots 2^{n-1} - 1$ (range split evenly, hence one extra negative). $n=5$: unsigned 0..31, signed $-16..15$ ($-16=-2^4$, $15=2^4-1$). **Bits needed:** store 1000 unsigned: $2^9=512 < 1000 \leq 2^{10}=1024 \Rightarrow 10$ bits; store -1000 signed: need $2^{n-1} \geq 1000 \Rightarrow n=11$ bits.

8. Prime factorization

Method: divide by smallest primes 2, 3, 5, 7, ... until quotient 1.
 $360=2\cdot 180=2^2\cdot 90=2^3\cdot 45=2^3\cdot 3\cdot 15=2^3\cdot 3^2\cdot 5$;
 $1000=2^3\cdot 5^3$.
Multiply via factorization (add exponents):
 $360\cdot 75$ ($75=3\cdot 5^2$):
 $2^3\cdot 3^2\cdot 5 \times 3\cdot 5^2=2^3\cdot 3^3\cdot 5^3=27,000$. **Check** $360\cdot 75=27,000$.

9–10. Expression trees & Lisp lists

Build tree: root = lowest-precedence operator (left-assoc: rightmost such); recurse into children.
 $a - b \div c + d \times e^f$: root $+$; left $a - b \div c$; right $d \times e^f$.



Tree→Lisp: node ($op\ L\ R$): $(+ (- a (/ b\ c)) (* d (^ e\ f)))$.
Lisp→infix: $(+ (* a\ b)\ c) \Rightarrow (a+b)\times c$. Custom precedence $f(a, b)=a \oplus b$: $f(f(x, y), z)$ parses as $x \oplus y \oplus z$; $f(x, f(y, z))$ as $x \oplus (f(y, z))$ (structure differs).

11–12. Transformations & typed rewrite rules

Substitution (replace a symbol/pattern everywhere): $x \rightarrow y+1$ in $x^2+2x+1 \Rightarrow (y+1)^2+2(y+1)+1$.

Expansion (distribute, then combine):
 $(2x+3)(x-1) \rightarrow 2x^2-2x+3x-3=2x^2+x-3$.

Typed rule $\frac{p:\text{int}}{q:\text{int}} \rightarrow \frac{p/g}{q/g}$ ($g=\text{gcd}$): $\frac{6}{8} \rightarrow \frac{3}{4}$
 applies, $\frac{x}{y}$ does not (x, y not ints).

Combine like terms ($c_1v+c_2v \rightarrow (c_1+c_2)v$ for variable v): $3x+5x \rightarrow 8x$; $3x+5y$ matches no rule.

13. Maxima (pseudocode only)

Add two base- B digit lists with carry: for $i:1$ thru n do ($s:=11[i]+12[i]+c$, $r[i]:\text{mod}(s, B)$, $c:\text{floor}(s/B)$) $\Rightarrow$ result list $[r]$ plus final carry c . Digits are kept in $[0, B-1]$ by mod/floor .

14–15. Modular arithmetic & notation

Reduce first, then operate:

$34+29 \equiv 6+1=7 \equiv 0 \pmod{7}$; $5\cdot 6 \equiv 30 \equiv 2 \pmod{7}$.

Exponentiate with repeated reduction (never the raw power): $3^4 \bmod 5$: $3^2=9 \equiv 4$, then $3^4 \equiv 4^2=16 \equiv 1$.

Negative input: $-17 \bmod 5$: add multiples of 5: $-17+20=3 \Rightarrow -17 \equiv 3$.

Notation: $[12]_7=[5]_7 \iff 12 \equiv 5 \pmod{7}$;

$2\cdot 3 \equiv 6 \pmod{7} \iff [2]_7[3]_7=[6]_7$.

16–17. Tables, inverses, zero divisors

Build the $\times$ -table mod n ; row a contains $1 \iff a$ invertible; row a has 0 in a nonzero column $\Rightarrow a$ zero divisor.

Mod 6 (composite): row 2 = 0, 2, 4, 0, 2, 4 (no 1; $2\cdot 3 \equiv 0$); row 3 = 0, 3, 0, 3, 0, 3; row 4 = 0, 4, 2, 0, 4, 2; row 5 = 0, 5, 4, 3, 2, 1 $\Rightarrow 5^{-1}=5$ ($5\cdot 5=25 \equiv 1$). Units: $\{1, 5\}$.

Mod 5 (prime, all nonzero invertible): row 3 has 1 at col 2 $\Rightarrow 3^{-1}=2$.

18. Polynomial modular arithmetic (worked)

Reduce mod $m(x)$ two ways: (A) use $m(x) \equiv 0$ to eliminate powers; (B) **long-divide** and keep the remainder (degree $< \deg m$). Mod x^2+1 : $x^2 \equiv -1$.
 (A) $2x^3+x^2-3x+4$: $2x^3=2x\cdot x^2 \equiv -2x$; $x^2 \equiv -1$; so $\equiv -2x-1-3x+4=-5x+3$ (class $[3-5x]$).
 (B) **long division** of x^3+2x^2+x+1 by x^2+1 :

$$\begin{array}{r} x^3+2x^2+x+1 \div (x^2+1) = x+2+\frac{-1}{x^2+1} \\ \underline{-x^3 -x} \\ 2x^2+1 \\ \underline{-2x^2 -2} \\ 3 \end{array}$$

$\Rightarrow$ quotient $x+2$, remainder -1 ; **check** $(x+2)(x^2+1)+(-1)=x^3+2x^2+x+1$.

Powers (reduce at the end):

$(1+x)^3=1+3x+3x^2+x^3 \equiv 1+3x-3-x=-2+2x$.
 Classes mod x^2+1 are all $a+bx$ (Gaussian integers $a+bi$).

19. Inverse existence / zero divisors (theorem)

Theorem: m invertible mod $n \iff \text{gcd}(m, n)=1$; n composite $\Rightarrow$ zero divisors exist.

5 mod 12: $\text{gcd}=1 \Rightarrow$ invertible; $5^{-1}=5$ ($5\cdot 5=25 \equiv 1$). 8 mod 12: $\text{gcd}=4 \neq 1 \Rightarrow$ **no** inverse; zero divisor: $8\cdot 3=24 \equiv 0$.
 Mod 10: units $\{1, 3, 7, 9\}$; $3^{-1}=7$ ($3\cdot 7=21 \equiv 1$); zero divisors $4\cdot 5=20 \equiv 0$, $6\cdot 5=30 \equiv 0$.

20–21. Groups: axioms & multiplication tables

Verify ALL 4 axioms (closure is the forgotten one). $(\mathbb{Z}_4, +)$: closure ($[a]+[b]=[a+b] \in \mathbb{Z}_4$), identity $[0]$, inverse $[a]+[-a]=[0]$ (e.g. $[1]+[3]=[0]$, $[2]+[2]=[0]$), associative (integers) $\Rightarrow$ group.
 $\mathbb{Z}_5^*=\{1, 2\}$: table: $2\cdot 2=4 \equiv 1 \Rightarrow$ rows $\{1, 2\}$, $\{2, 1\}$;
 $1^{-1}=1$, $2^{-1}=2$; order-2 group.
 $\mathbb{Z}_4^*=\{1, 3\}$: $3\cdot 3=9 \equiv 1 \Rightarrow$ group. $\{1, 2, 3\}$ with $\times$ mod 4 is **not**: $2\cdot 2=0 \notin$ set (closure fails).

22–23. Order, generator, Lagrange

Order: smallest $n>0$ with $g^n=e$. **Compute powers until e:** in $\mathbb{Z}_7^*$: $2^1=2$, $2^2=4$, $2^3=8 \equiv 1 \Rightarrow \text{ord}(2)=3$.

Generator $\iff \text{ord}(g)=|G|$: 3: powers 3, 2, 6, 4, 5, 1 (6 distinct) $\Rightarrow$ generator of $\mathbb{Z}_7^*$.
 $\mathbb{Z}_{11}^*$ ($|G|=10$): 2: 2, 4, 8, 5, 10, 9, 7, 3, 6, 1 $\Rightarrow \text{ord}=10 \Rightarrow$ generator; 3: 3, 9, 5, 4, 1 $\Rightarrow \text{ord}=5$.

Lagrange: order divides $|G|$: $|\mathbb{Z}_{12}^*|=|\varphi(12)|=4 \Rightarrow$ orders $\in \{1, 2, 4\}$; $\varphi(9)=6 \Rightarrow \{1, 2, 3, 6\}$.

24. Fermat big powers

Method: (1) check p prime and $\text{gcd}(a, p)=1$; (2) $a^{p-1} \equiv 1 \pmod{p}$; (3) **reduce base mod p** ; (4) $a^{(p-1)q+c} \equiv a^c$.
 $3^{100} \bmod 7$: 7 prime, $7 \nmid 3$; $3^6 \equiv 1$; $100=6\cdot 16+4$; $3^{100} \equiv 3^4=81 \equiv 4$.
 $5^{100} \bmod 13$: $5^{12} \equiv 1$; $100=12\cdot 8+4$; $5^4=625 \equiv 1$.
 $2^{1000} \bmod 7$: $1000=6\cdot 166+4$; $2^4=16 \equiv 2$.

25. Bases 16/8 (hex/octal) shortcuts

Dec→hex: same $\div 16$ method, digits 0..9, A..F:
 $45=2\cdot 16+13 \rightarrow 0x2D$. **Check:** $2\cdot 16+13=45$.

Hex→bin: one hex digit = 4 bits:

$2D \rightarrow 0010\ 1011=1011012$;

$11001012 \rightarrow 0011\ 1001 \rightarrow 0x39=57$.

Hex→dec: $0x2D=2\cdot 16+13=45$. Group bits in 4s (hex) or 3s (octal) from the right.

26. IEEE 754 special values

Single (32-bit, bias 127): smallest positive normal $2^{-126} \approx 1.17549 \times 10^{-38}$; largest normal $(2^{-2}-2^{23})2^{127} \approx 3.40282 \times 10^{38}$.

24-bit float (bias 127, 7-bit exp, 16-bit mantissa):
 $E=e+127$; 16-bit course format = bias 24, 5-bit exp, 10-bit mantissa (see §35).

27. LCM, more Euclid, base-B practice

$\text{lcm}(a, b) = \frac{ab}{\text{gcd}(a, b)}$: $\text{lcm}(12, 18) = \frac{216}{6} = 36$.

Euclid again: $\text{gcd}(84, 30)$: $84=30\cdot 2+24$,
 $30=24\cdot 1+6$, $24=6\cdot 4+0 \Rightarrow \text{gcd}=6$.
 Base-60: $3601=[1, 0, 1]_{60}$ (60^2+1), $125=[0, 2, 5]_{60}$;
 sum $[1, 2, 6]_{60}=3600+120+6=3726$ ($3601+125$).

28. More trees & rewrites

$a\cdot(b+c) - d$: root $-$, left $\times$, right d ; Lisp: $(- (* a (+ b\ c))\ d)$.

Substitute $z=x+y$ in z^2-1 : $\Rightarrow (x+y)^2-1$; expand $(x+y)^2=x^2+2xy+y^2$.

Rule $c_1v^2+c_2v^2 \rightarrow (c_1+c_2)v^2$: $2x^2+3x^2 \rightarrow 5x^2$;
 $2x^2+3y^2$ no match ($x \neq y$).

29. Solving simple linear congruences

Solve $ax \equiv b \pmod{n}$: if $\text{gcd}(a, n) \nmid b$, **no solution**; else divide all three by gcd , multiply by a^{-1} .

$3x \equiv 1 \pmod{7}$: $\text{gcd}(3, 7)=1$; $3^{-1}=5$ ($3\cdot 5=15 \equiv 1$);
 $x \equiv 5 \pmod{7}$.

$4x \equiv 2 \pmod{6}$: $\text{gcd}(4, 6)=2$ divides 2 $\Rightarrow$ 2

solutions; divide: $2x \equiv 1 \pmod{3}$; $2^{-1}=2$; $x \equiv 2 \pmod{3} \Rightarrow x=2, 5$ (check $4\cdot 2=8 \equiv 2$, $4\cdot 5=20 \equiv 2$).

30. More generators & orders

$\mathbb{Z}_{13}^*$ ($|G|=12$): powers of 2:
 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, 1 $\Rightarrow \text{ord}(2)=12 \Rightarrow$ generator.

$\mathbb{Z}_7^*$: $\text{ord}(2)=3$, $\text{ord}(3)=\text{ord}(5)=6$, $\text{ord}(4)=3$,
 $\text{ord}(6)=2$; generators = $\{3, 5\}$.

$\mathbb{Z}_{11}^*$ ($|G|=10$): generators = $\{2, 6, 7, 8\}$. $\mathbb{Z}_8^+$:
 $[1], [3], [5], [7]$ order 8; $[2], [6]$ order 4; $[4]$ order 2.

Cyclic $\iff \exists g$ with $\text{ord}(g)=|G|$.

31. More Fermat practice

$3^{49} \bmod 7$: $3^6 \equiv 1$; $49=6\cdot 8+1$; $3^{49} \equiv 3^1=3$.
 $11^{1000} \bmod 17$: $11^{16} \equiv 1$; $1000=16\cdot 62+8$;
 $11^8=(11^2)^4=(121 \equiv 2)^4=16 \equiv -1 \Rightarrow 11^{1000} \equiv 16$.
 $2^{200} \bmod 31$: $2^{30} \equiv 1$; $200=30\cdot 6+20$;
 $2^{20}=(2^5)^4=32^4 \equiv 1$. **Shortcut** if base $\equiv 0$:
 $21^{50} \bmod 7=0$.

Mod-6 $\times$ table (read off inverses & zero divisors):

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

5) $\Rightarrow 5^{-1}=5$; rows 2,3,4 hit 0 $\Rightarrow$ zero divisors. Units $\{1, 5\}$.

32. Signed overflow (two's complement)

Add two positives and get a negative (or vice-versa) $\Rightarrow$ overflow. 8-bit: $127+1$:
 $01111111+00000001=10000000=-128$ (overflow!);
 $100+100=200 > 127$ also overflows. **Trap:** -128 has no positive twin.

33. Constructing the D_3 table

Relations: $r^3=e$, $\ell_i^2=e$; $rol_1=\ell_2$, $r^2ol_1=\ell_3$; entry = **column first, then row**.

Row r : $roe=r$, $ror=r^2$, $ror^2=e$, $rol_1=\ell_2$, $rol_2=\ell_3$, $rol_3=\ell_1$; **check:** each row is a permutation of all 6 (Latin square). $r^{-1}=r^2$, $\ell_i^{-1}=\ell_i$.

34. Constructing a $+$ table mod n

Row a of the $+$ -table = row 1 shifted cyclically by a : mod 7, row 1 = 0, 1, 2, 3, 4, 5, 6; row 3 = 3, 4, 5, 6, 0, 1, 2. $[a]+[b]=[a+b \bmod n]$.
Additive inverse $[-a]=[n-a]$: mod 7,
 $[3]^{-1}=[4]$; mod 6, $[2]^{-1}=[4]$, $[5]^{-1}=[1]$.

35. IEEE 754 16-bit (course: bias 24) worked

Convert 2.5 (5-bit exp, 10-bit mantissa, bias 24):
 $2.5=10.1_2=1.01_2 \times 2^1$; $s=0$; $f=01$ padded to 10 bits;
 $E=1+24=25=11001_2 \Rightarrow$ bits
 $0\ 11001\ 0100000000$.

Decode: $E=25 \Rightarrow e=1$; $(1.01)_2=1.25$; $1.25 \times 2=2.5$.
Check: $\text{max}=(2-2^{-10})2^{31-24}=255.875$.

36. APN multiplication (schoolbook)

Multiply digit lists base B : **each digit pair $a_i b_j$ adds to position $i+j$** , carrying down.
 $[4, 2] \times [3, 1]$ base 10 ($24 \times 13=312$): units $4\cdot 3=12 \rightarrow$ digit 2, carry 1; then $4\cdot 1+2\cdot 3+1=11 \rightarrow$ digit 1, carry 1; then $2\cdot 1+1=3$. Result $[2, 1, 3]=312$.

Quick Reference: Powers of 2: $2^{10}=1024$, $2^{11}=2048$, $2^{20} \approx 10^6$, <